simplifying the computation to

$$\mathcal{L}^{\dagger}_{2,[\![\mathrm{OYZYO}]\!]}\rho_\beta = \frac{1}{12}\left[\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Z}},\mathcal{L}^{\dagger}_{\mathrm{Y}}\right]\rho_\beta - \frac{1}{24}[\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Y}},\mathcal{L}^{\dagger}_{\mathrm{Z}}]\rho_\beta. \tag{7.29}$$

The inhomogeneity for the schemes $[\![\mathrm{OABAO}]\!]$ and $[\![\mathrm{OBABO}]\!]$ is hence a combination of the two commutator terms

$$\begin{aligned}\left[\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{A}},\mathcal{L}^{\dagger}_{\mathrm{B}}\right]\rho_\beta &= \left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \left[\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{B}},\mathcal{L}^{\dagger}_{\mathrm{A}}\right]\rho_\beta &= -\left(2\beta \boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta,\end{aligned}$$

giving overall equations to solve

$$\begin{aligned}\mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OABAO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OABAO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)\rho_\beta,\\ \mathcal{L}^{\dagger}_{\mathrm{LD}}\left(\rho_\beta f_{2,[\![\mathrm{OBABO}]\!]}\right) &= -\mathcal{L}^{\dagger}_{2,[\![\mathrm{OBABO}]\!]}\rho_\beta\\ &= \frac{\beta}{24}\left(2\nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p} - 6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U\right)\rho_\beta.\end{aligned}$$

We can see immediately that the Fredholm alternative holds in both cases, as both equations have right-hand sides that are of odd order in p, and hence will average to 0.

If we instead consider schemes which use a central Ornstein-Uhlenbeck update, then the inhomogeneous term becomes significantly more complex, as we no longer have a significant number of cancellations in $\mathcal{L}^{\dagger}_2\rho_\beta$. Instead, we shall avoid computing the right-hand side for these schemes altogether, in favor of utilizing the following lemma [223]:

Lemma 7.1 *Consider two numerical schemes with associated operators governing the evolution of measure $\mathcal{ST}$ and $\mathcal{TS}$, with unique associated invariant distributions $\hat{\rho}_{ST}$ and $\hat{\rho}_{TS}$ respectively, such that*

$$\mathcal{ST}\hat{\rho}_{ST} = \hat{\rho}_{ST}, \qquad \mathcal{TS}\hat{\rho}_{TS} = \hat{\rho}_{TS}.$$

Under suitable ergodicity assumptions (such that in the long-time limit, all distributions converge to the invariant distribution), we have

$$\hat{\rho}_{ST} = \mathcal{S}\hat{\rho}_{TS}, \quad \textit{and} \quad \hat{\rho}_{TS} = \mathcal{T}\hat{\rho}_{ST}.$$